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GEN AI ARCHITECT JOURNEY · PHASE-003

WhatsApp Automation Bot

Consent-based messaging, monitored and reported

Track	Playwright - browser automation
What it does	Messages, watches for replies, reports the outcome
Single file	Playwright-WhatsApp-Automation-Bot.py
Needs	Microsoft Edge · a WhatsApp account · openpyxl
Folder	Phase-003-Robotic-Process-Automation-(RPA)/Playwright Automation/Playwright Assignment

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1. What it does

Sends personalised, consent-based WhatsApp Web messages through Microsoft Edge, monitors for replies, and generates dated JSON and Excel reports.

Consent is a design constraint, not a footnote

Use this only with people who have explicitly agreed to receive the messages. Automated or bulk messaging may violate WhatsApp's terms and can get an account restricted.

- Opens WhatsApp Web in Edge with a reusable local browser profile.
- Supports manual QR login and manual **Use here** confirmation.
- Reads **Name**, **Phone**, and optional **Message** values from **contacts.xlsx**.
- Searches each international number and continues when one contact fails.
- Reads the displayed chat name and personalises **{name}** placeholders.
- Extracts the latest three incoming messages and saves a final screenshot.
- Masks phone numbers in every log and generated report.

2. The message flow

Time	What the bot does
0s	Sends the personalised opening message
0-300s	Watches for a new incoming reply
180s	Sends exactly one follow-up, only if no reply has arrived
on reply	Sends a confirmation immediately
300s	Records no_reply and moves to the next contact

TEXT

Hi {name}! This is a test message from my Playwright WhatsApp Automation project. Please reply "Hi". Thank you!

-- follow-up at 180s, only when no reply has arrived --

Hi {name}, just checking in. Please reply "Hi" when available.

-- on a valid reply --

Thank you {name}! Your reply has been received successfully.

3. Setup and run

POWERSHELL

```
cd "Playwright Automation\Playwright Assignment"
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install playwright openpyxl

python Playwright-WhatsApp-Automation-Bot.py
```



The script launches the installed Microsoft Edge channel, so a separate Chromium download is normally unnecessary.

4. The input workbook

contacts.xlsx supplies the recipients. It is deliberately **not committed** - it holds real, unmasked phone numbers. Create it locally with these columns:

Column	Required	Contents
Name	Yes	Display name, used for {name} personalisation
Phone	Yes	Full international number, for example +91XXXXXXXXXX
Message	No	Overrides the default opening message

5. Privacy and what is excluded

This is the one project in the phase that touches real personal data, so the boundaries are drawn explicitly.

Committed	Never committed
The bot source and its documentation	contacts.xlsx - real, unmasked phone numbers
The project brief	.whatsapp_profile/ - a live logged-in session
Development notes (the Error Fix images)	output/ - conversation reports and screenshots

Masking is built in

The bot masks numbers in its own output: a generated report shows **+91*****30**, never the full number. The exclusions above cover the input side, which the bot cannot mask for you.

Next

This is the final project of Phase-003. Phase-004 moves from driving other people's software to building your own interfaces.

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